Task 3.1: Implementing Dynamic Features with JavaScript

Overview

In this Touchstone task, you will begin adding simple dynamic behavior to your webpages using JavaScript. You will use click events to display alert messages when users interact with different buttons and form elements. These features introduce the basic JavaScript skills that you will continue developing in later Touchstone tasks.

You will complete the following:

- Add a JavaScript alert for the **Subscribe** button in the footer of every webpage.
- On the **Gallery** page, add alert messages to the **Add to Cart**, **Clear Cart**, and **Process Order** buttons. These buttons act as **placeholders** for features you will build later. For this task, each button only needs to display a simple alert when clicked.
- On the **About Us/Contact** page, add a JavaScript alert to the **Submit** button on the contact form.

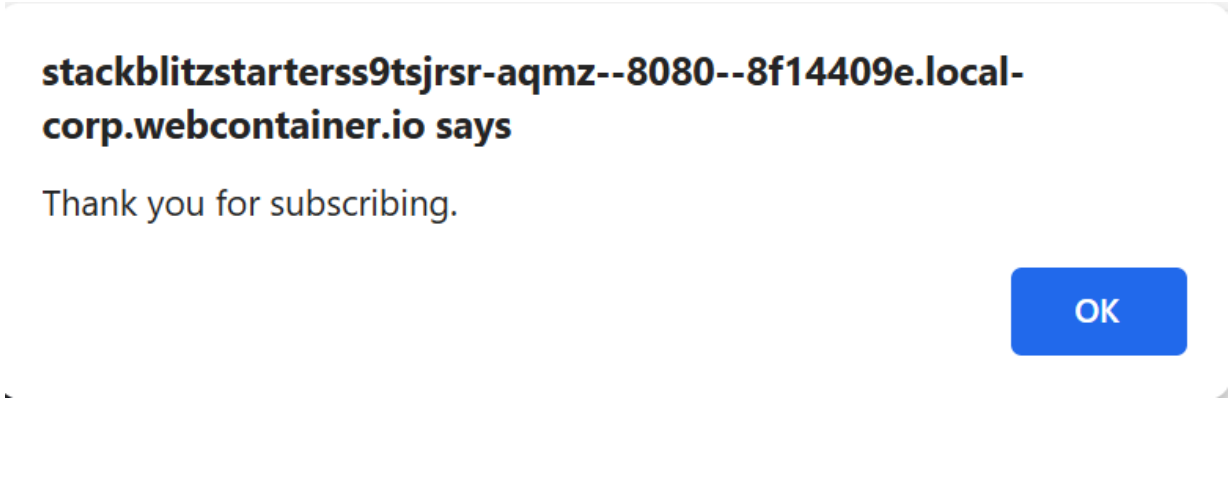
Each alert should activate through a click event and display one of the following messages:
• “Thank you for subscribing.”
• “Item added to the cart.”
• “Cart cleared.”
• “Thank you for your order.”
• “Thank you for your message.”

Finally, document any issues or errors you encountered while developing your code. Remember that any component colors, font colors, and typography styling should be consistent with your choices in Task 1 and should be appropriate according to the client description provided in the **Client Scenario and Criteria**.

Instructions

Provide your responses in the following table.

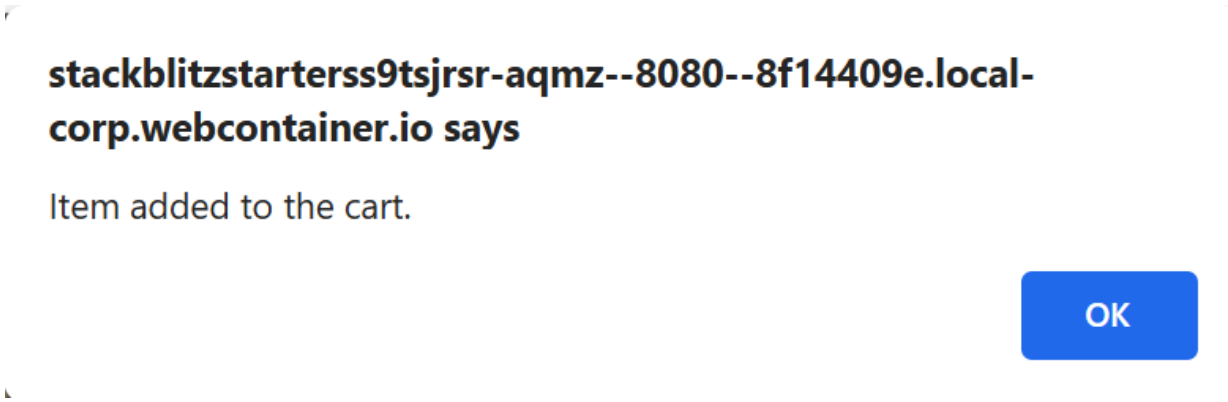
Rendered Subscribe Feature Alert Window Screenshot *Hint: Provide a screenshot showing the alert message that appears when the user clicks the Subscribe button.*



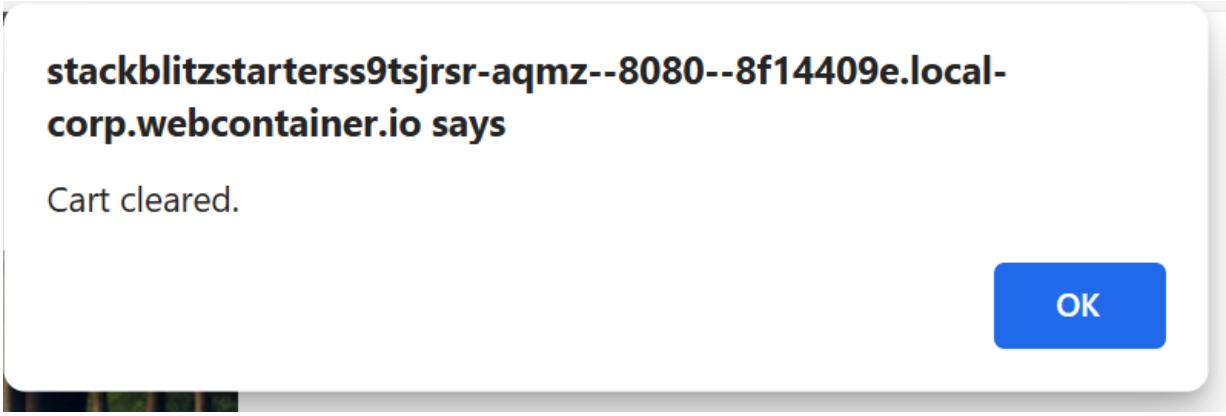
Subscribe Feature Alert Window Analysis
Hint: Briefly describe how you implemented the Subscribe button alert.

I added a click event listener to the Subscribe button in the footer. When the button is clicked, the event handler triggers an `alert("Thank you for subscribing.")`. The alert displays correctly on every page because the script file is linked across the site.

Rendered Add to Cart Button Alert Screenshot (Gallery Page)
Hint: Provide a screenshot showing the alert message that appears when clicking any "Add to Cart" button on the Gallery page.



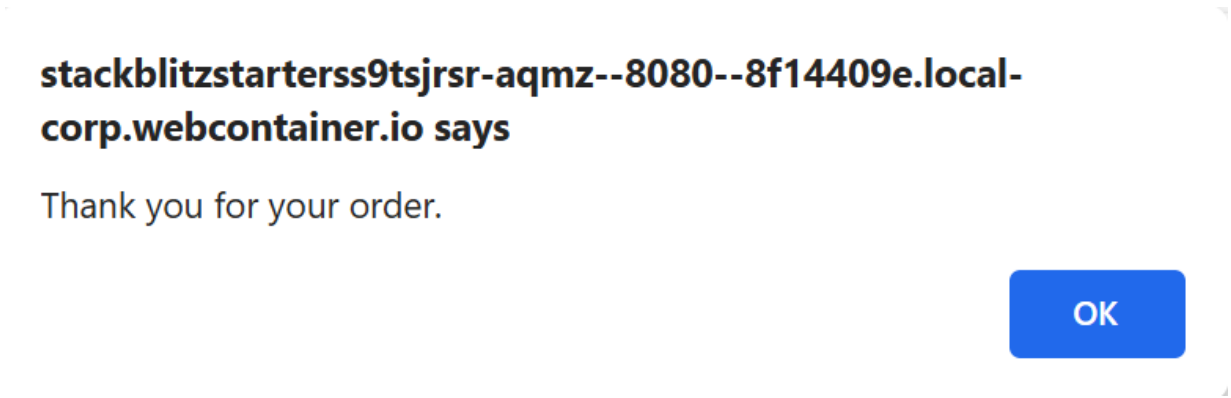
Add to Cart Button Analysis *Hint: Describe how JavaScript is applied to the Add to Cart button. Remember: no shopping cart logic is required.*
I selected every “Add to Cart” button on the Gallery page using `document.querySelectorAll`. Then, I added a click event listener to each button so that clicking any item shows the alert message “Item added to the cart.” No cart logic was required for this step.

Rendered Clear Cart Button Alert Screenshot (Gallery Page) *Hint: Provide a screenshot showing the alert message that appears when clicking the “Clear Cart” button.*


Clear Cart Button Analysis
Hint: Describe how the alert message for the Clear Cart button is triggered.

The Clear Cart button has a simple click event listener that calls `alert("Cart cleared.")` whenever the button is pressed. This prepares the button for future cart functionality that will be added in later lessons.

Rendered Process Order Button Alert Screenshot (Gallery Page)
Hint: Provide a screenshot showing the alert message that appears when clicking the "Process Order" button.



Process Order Button Analysis
Hint: Explain how you added the alert to the Process Order button.

To create this alert, I added an event listener to the Process Order button. When clicked, it runs a small function that displays the message "Thank you for your order." This confirms the button's event handler is working.

Rendered About Us/Contact Page Form Submission Alert Window Screenshot
Hint: Provide a screenshot showing the alert that appears when the Submit button is clicked on the Contact form.

About Us/Contact Page Analysis

Hint: Describe how JavaScript is applied to the Contact page's Submit button.

On the Contact page, I assigned a click event listener to the Submit button. When the user clicks Submit, the event handler prevents a page reload and shows an alert saying "Thank you for your message." This verifies the form submission interaction.

Rendered About Us/Contact Page Input Validation Screenshot

Hint: Provide a screenshot showing any validation messages that appear if required fields are empty when the form is submitted. These messages are generated automatically by the browser through built-in HTML form validation. You do not need to write any JavaScript validation for this step.

About Us/Contact Page Input Validation Analysis

Hint: Briefly describe what happens when the form is submitted with missing information. The validation messages you see are created by the browser because the form fields use HTML attributes such as required. No additional JavaScript is needed to produce these message.
If the required fields are empty when the user clicks Submit, the browser automatically shows a validation message such as “Please fill out this field.” These messages come from HTML form attributes like `required`, so no JavaScript is needed to make them appear.

Testing and Error Reporting *Hint: Record any errors you encountered while writing or testing your JavaScript code. Describe how you attempted to fix them. If everything worked as expected, note that here.*
Record the details of the error from the IDE here. A screenshot or copy-and-paste of the text into this entry is acceptable. At first, my script gave an error saying one of the buttons was null. This happened because the script loaded before the page elements were created.
Describe what you attempted in order to fix the error. I moved my `<script>` tag to the bottom of the HTML `<body>`. After that, the browser loaded the page before running the JavaScript, and all my event listeners worked correctly.